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Lab #1 — VMware Installation and Virtual Machine Setup

Course: System & Network Administration / Cloud Computing

Lab Number: 01_VMware Lab

Topic: VMware Installation, VM Creation and Linux Environment Setup

Tools Required: VMware Workstation, Ubuntu/Linux ISO, Windows Host Machine

Lab Learning Outcomes

- Understand virtualization concepts
- Install VMware Workstation
- Create and configure virtual machines
- Install Linux operating system
- Manage virtual hardware resources
- Perform basic verification and VM operations

Part A — VMware Installation

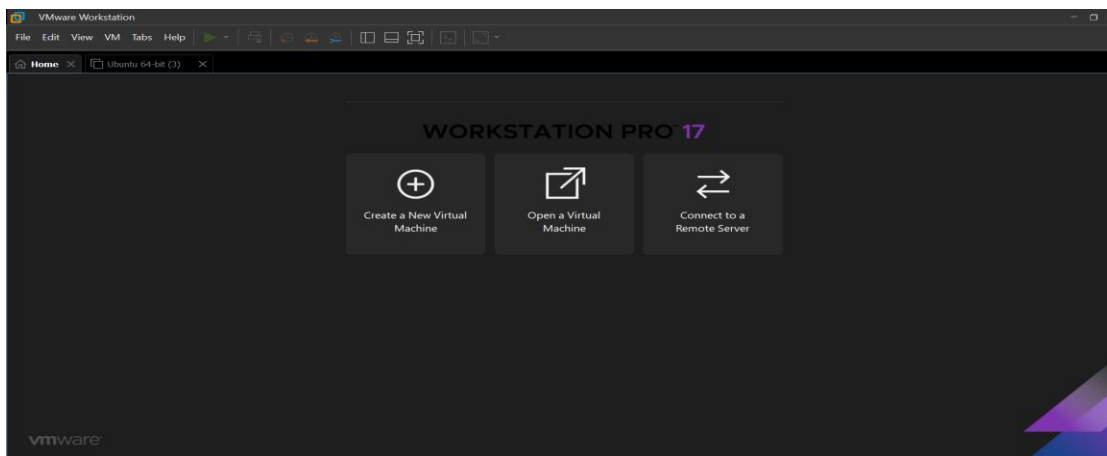
Step 1: Download VMware Workstation Player/Pro from <https://www.vmware.com>

Step 2: Run the installer and accept license agreement

Step 3: Select default installation components

Step 4: Choose installation location

Step 5: Complete installation and restart system if required



Part B — Create a Virtual Machine

Step 1: Open VMware Workstation

Step 2: Click Create a New Virtual Machine

Step 3: Select Typical Installation

Step 4: Browse and attach Linux ISO file

Step 5: Choose Guest Operating System (Linux/Ubuntu)

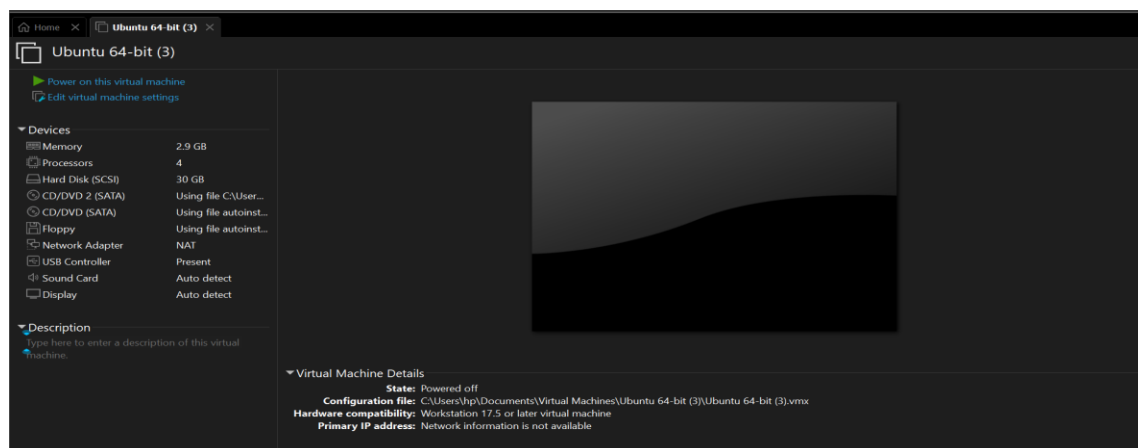
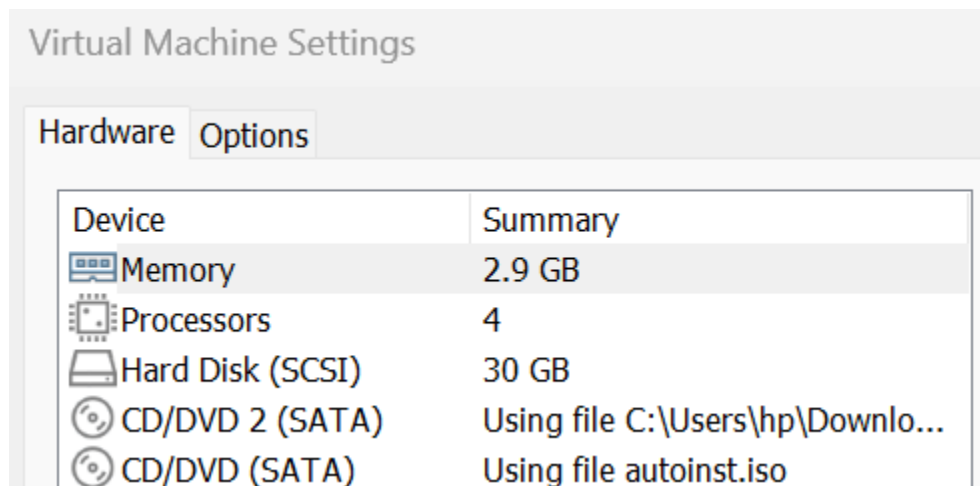
Step 6: Assign virtual machine name and storage location

Step 7: Allocate RAM (Minimum 2GB recommended)

Step 8: Allocate CPU cores

Step 9: Create virtual disk (Minimum 20GB)

Step 10: Finish VM creation



Part C — Configure Virtual Machine

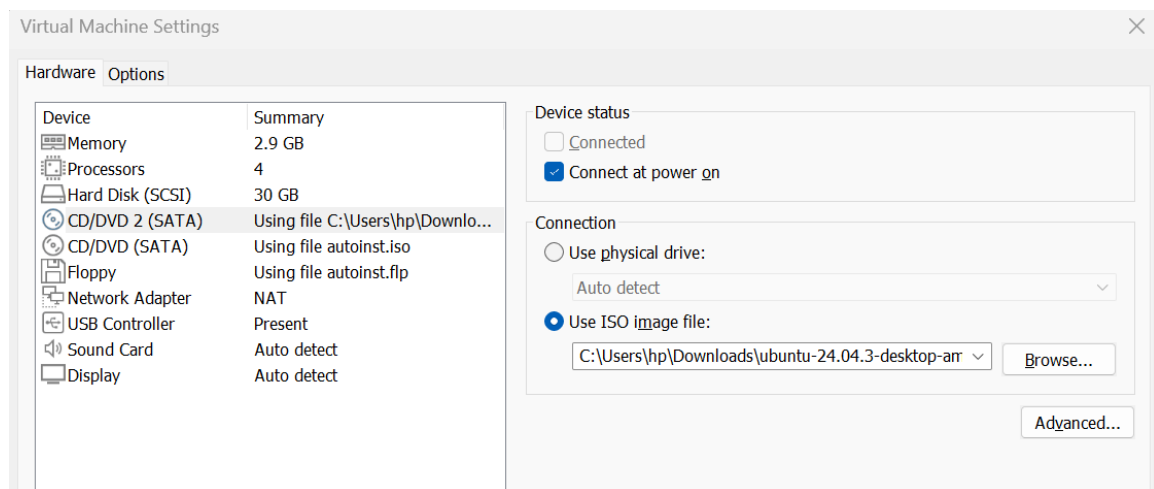
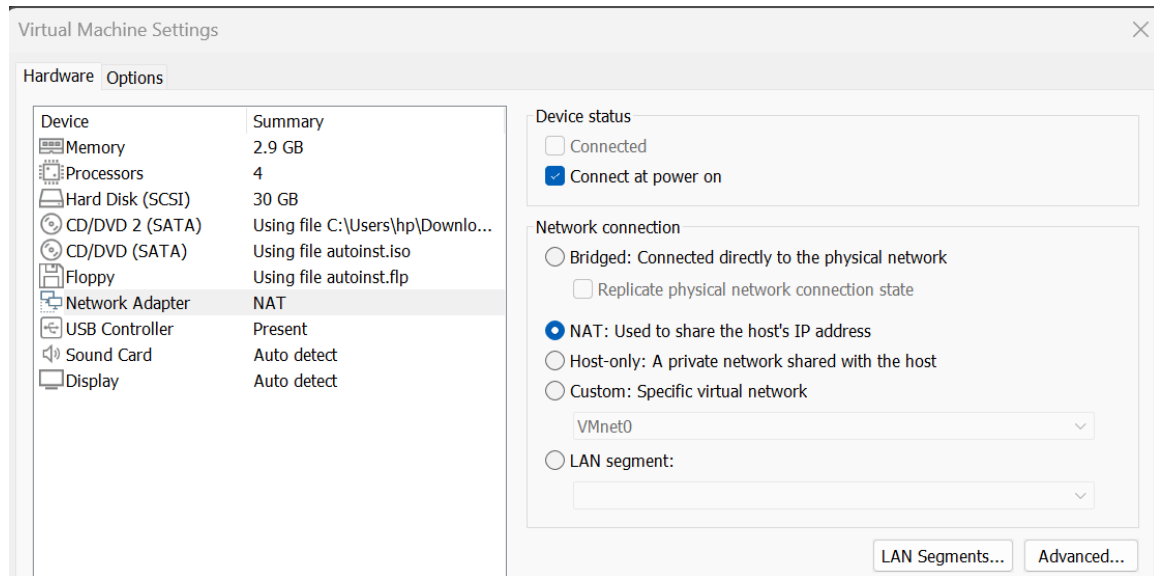
Step 1: Open VM Settings

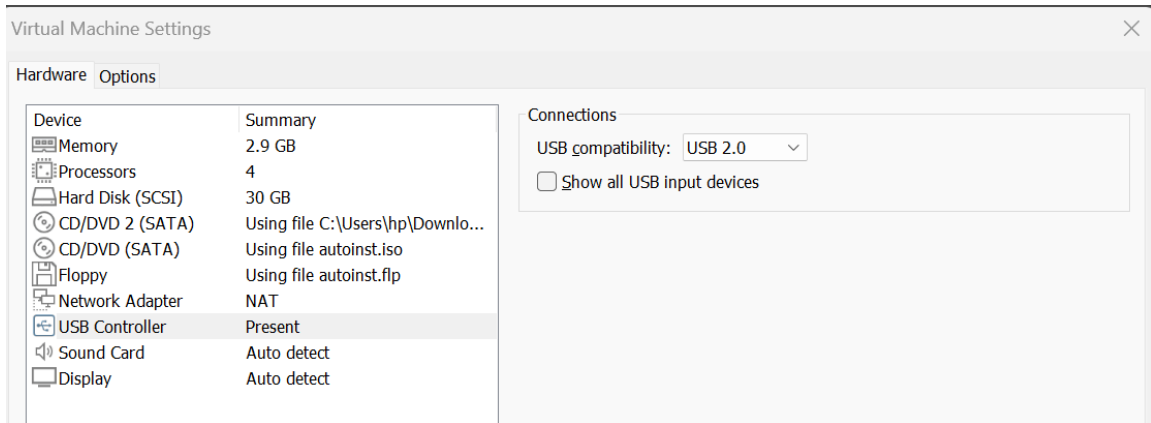
Step 2: Adjust RAM and CPU settings

Step 3: Configure Network Adapter (NAT/Bridged)

Step 4: Enable USB Controller

Step 5: Verify storage and ISO attachment





Part D — Install Linux Operating System

Step 1: Start the Virtual Machine

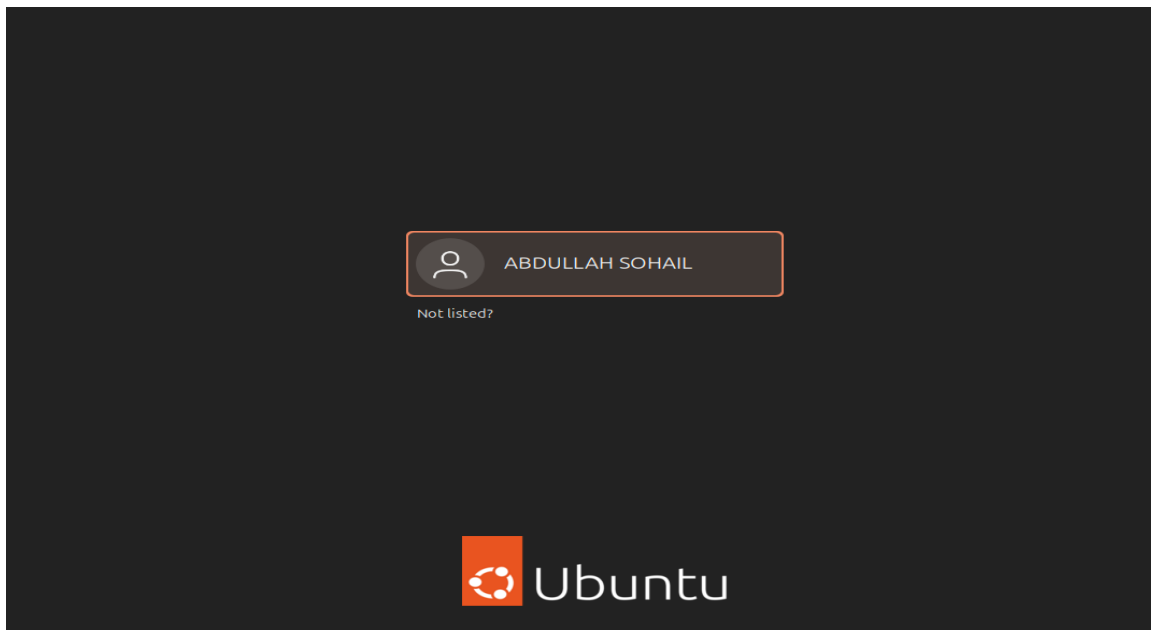
Step 2: Select Install Ubuntu/Linux option

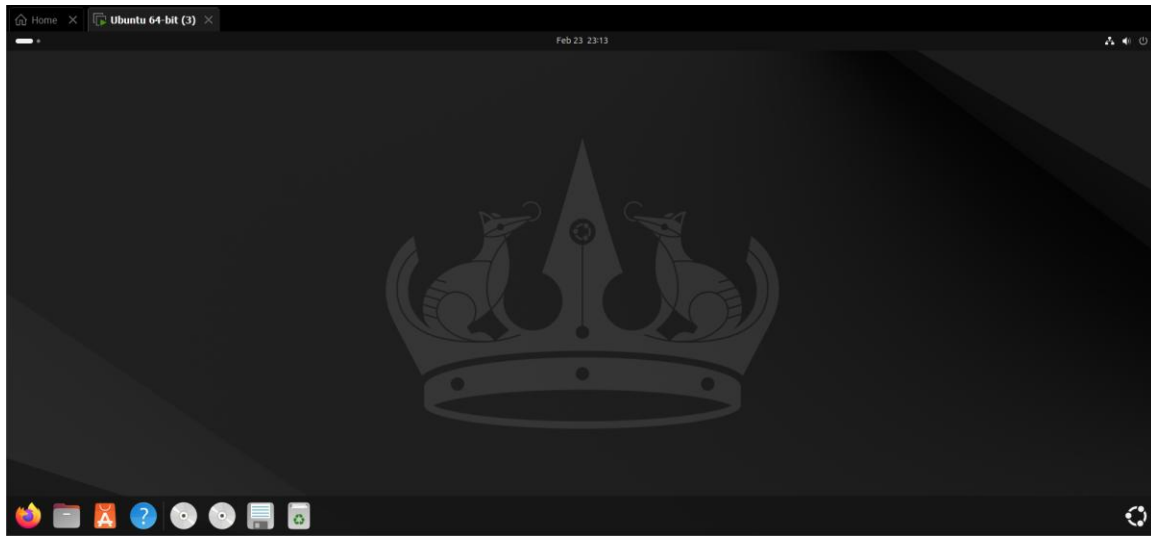
Step 3: Choose language and keyboard layout

Step 4: Create user account and password

Step 5: Select automatic disk partitioning

Step 6: Complete installation and reboot VM





Part E — Verify Installation

Step 1: Login to Linux system

Step 2: Open Terminal

Step 3: Check current user using whoami command

Step 4: Check system information using uname -a

Step 5: Verify disk using df -h

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ whoami
abdullah-sohail
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ uname -a
Linux abdullah-sohail-VMware-Virtual-Platform 6.17.0-14-generic #14-24.04.1-Ubuntu SMP PREEMPT_DYNAMIC Thu Jan 15 15:52:10 UTC 2 x86_64 x86_64 x86_64 GNU/Linux
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            286M  1.7M  284M   1% /run
/dev/sda2        30G   8.8G   20G  32% /
tmpfs            1.4G   0  1.4G   0% /dev/shm
tmpfs            5.0M   8.0K   5.0M   1% /run/lock
tmpfs            286M  116K  286M   1% /run/user/1000
/dev/sr0         96M   96M   0 100% /media/abdullah-sohail/CDROM
/dev/sr1         6.0G   6.0G   0 100% /media/abdullah-sohail/Ubuntu 24.04.3 LTS amd64
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

Practice Tasks (Student Activities)

Task 1: Create a new folder named vm_lab inside Home directory.

Task 2: Create three text files inside vm_lab.

Task 3: Display current directory path.

Task 4: List all files including hidden files.

Task 5: Rename one file.

Task 6: Copy one file to another directory.

Task 7: Delete one file.

Task 8: Create subfolder inside vm_lab.

Task 9: Move a file into subfolder.

Task 10: Check system processes using top command.

Task 11: Check network configuration using ip a.

Task 12: Shutdown virtual machine using command line.

Task 13: Restart virtual machine.

Task 14: Take VMware snapshot.

Task 15: Restore VM using snapshot.

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ cd ~
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ mkdir vm_lab
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls
Desktop Documents Downloads Music Pictures Public snap Templates Videos vm_lab
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ cd vm_lab
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~/vm_lab$ touch file1.txt
touch file2.txt
touch file3.txt
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~/vm_lab$ ls
file1.txt file2.txt file3.txt
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~/vm_lab$ pwd
/home/abdullah-sohail/vm_lab
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~/vm_lab$ ls -la
.  ..  file1.txt file2.txt file3.txt
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~/vm_lab$ mv file1.txt newfile.txt
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~/vm_lab$ ls
file2.txt file3.txt newfile.txt
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~/vm_lab$ cd ~
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ cp vm_lab/file2.txt .
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls
Desktop Documents Downloads file2.txt Music Pictures Public snap Templates Videos vm_lab
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ rm vm_lab/file3.txt
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls vm_lab
file2.txt newfile.txt
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ mkdir vm_lab/subfolder
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls vm_lab
file2.txt newfile.txt subfolder
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ mv vm_lab/newfile.txt vm_lab/subfolder/
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls vm_lab/subfolder
newfile.txt
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ top

top - 23:19:58 up 8 min,  1 user,  load average: 0.11, 0.40, 0.34
Tasks: 310 total,  1 running, 309 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.2 us,  0.2 sy,  0.0 ni, 99.6 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 2852.6 total,  488.8 free, 1150.0 used, 1411.4 buff/cache
MiB Swap: 2852.0 total, 2852.0 free,  0.0 used. 1702.7 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM    TIME+  COMMAND
 3599 abdulla+  20   0  14524   5972  3744  R   1.0   0.2   0:00.28 top
   738 root      20   0  244756  9536  8076  S   0.7   0.3   0:01.86 vmtotlsd
 2508 abdulla+  20   0 4184872 265624 134932 S   0.7   9.1   0:23.34 gnome-shell
 2238 root      20   0      0      0      0  I   0.3   0.0   0:00.66 kworker/0:4-events
    1 root      20   0  23276  14644  9852  S   0.0   0.5   0:05.83 systemd
    2 root      20   0      0      0      0  S   0.0   0.0   0:00.07 kthreadd
    3 root      20   0      0      0      0  S   0.0   0.0   0:00.00 pool_workqueue_release
    4 root      0 -20      0      0      0  I   0.0   0.0   0:00.00 kworker/R-rcu_gp
    5 root      0 -20      0      0      0  I   0.0   0.0   0:00.00 kworker/R-sync_wq
    6 root      0 -20      0      0      0  I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
    7 root      0 -20      0      0      0  I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
    8 root      0 -20      0      0      0  I   0.0   0.0   0:00.00 kworker/R-netns
    9 root      20   0      0      0      0  I   0.0   0.0   0:00.00 kworker/0:0-events
   11 root      0 -20      0      0      0  I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
   12 root      20   0      0      0      0  I   0.0   0.0   0:00.00 kworker/u512:0-ipv6_addrconf
   13 root      0 -20      0      0      0  I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
   14 root      20   0      0      0      0  S   0.0   0.0   0:00.09 ksoftirqd/0
   15 root      20   0      0      0      0  I   0.0   0.0   0:01.14 rcu_preempt
   16 root      20   0      0      0      0  S   0.0   0.0   0:00.06 rcu_exp_par_gp_kthread_worker/1

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:54:7b:88 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.159.131/24 brd 192.168.159.255 scope global dynamic noprefixroute ens33
        valid_lft 1241sec preferred_lft 1241sec
    inet6 fe80::20c:29ff:fe54:7b88/64 scope link
        valid_lft forever preferred_lft forever
```

LAB REPORT:

I learned about virtualization and installed VMware Workstation on my Windows system. I successfully created a new virtual machine and configured hardware settings such as RAM, CPU, and network (NAT mode). After that, I installed the Ubuntu Linux operating system inside the virtual machine. I verified the installation using terminal commands like `whoami`, `uname -a`, and `df -h`. Finally, I performed basic Linux file management tasks and created a snapshot of the virtual machine. This lab helped me understand virtual machines and Linux environment setup.

Submission Requirements

- Screenshots of VMware installation
- Screenshot of created virtual machine
- Screenshot of Linux terminal verification commands
- Short lab report (5–6 lines)